

MATHEMATICS 2

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Mathematics 2

- 1 The sum of the first 3 terms of a geometric progression is 8. The sum of the first 6 terms of the same progression is 72.
What is the common ratio of the progression?

A 2
B 3
C 4
D 8
E 9

- 2 The diagram shows a triangle with side lengths $x + 3$, $2x - 1$, and $11 - x$.
Find the complete set of real values of x for which a non-degenerate triangle with these side lengths exists.

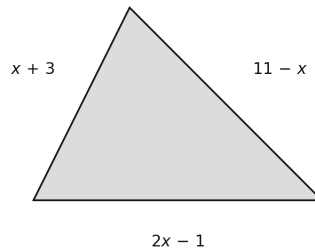


Diagram not to scale

A $x > \frac{9}{4}$
B $x < \frac{15}{2}$
C $\frac{1}{2} < x < 11$
D $\frac{9}{4} < x < \frac{15}{2}$
E $\frac{9}{4} < x < 11$
F $\frac{1}{2} < x < \frac{15}{2}$
G $\frac{1}{2} < x < \frac{9}{4}$
H $\frac{15}{2} < x < 11$

- 3 The diagram shows a regular hexagon $ABCDEF$ with sides of length a .

A circular arc with centre A is drawn from vertex C to vertex E .

What is the area of the shaded region bounded by the line segments CD and DE and the arc CE ?

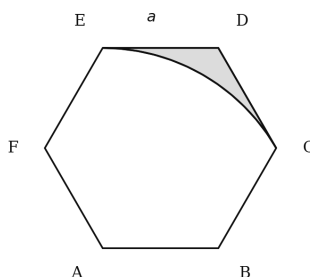


Diagram not to scale

- A $\left(\frac{\sqrt{3}-\pi}{2}\right)a^2$
- B $\left(\frac{2\sqrt{3}-\pi}{4}\right)a^2$
- C $\left(\frac{2\sqrt{3}-\pi}{2}\right)a^2$
- D $\left(\frac{3\sqrt{3}-\pi}{2}\right)a^2$
- E $\left(\frac{4\sqrt{3}-\pi}{4}\right)a^2$
- F $\left(\frac{3\sqrt{3}-2\pi}{4}\right)a^2$

- 4 What is the value of the definite integral

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (3 + \cos x + x \cos x) \, dx$$

- A 2π
- B 3π
- C $3\pi + 1$
- D $3\pi + 2$
- E 4π

- 5 A parabola has the equation $y = x(k - x)$ for some constant $k > 0$.
A triangle is formed by the two points where the parabola meets the x -axis and by the vertex of the parabola.

What is the total area of the regions that lie between the parabolic arc and the sides of the triangle?

- A $\frac{k^3}{24}$
B $\frac{k^3}{12}$
C $\frac{k^3}{8}$
D $\frac{k^3}{6}$
E $\frac{k^3}{4}$

- 6 Find the area of the finite region defined by the inequalities:

$$y \geq |2x + 4|$$

$$y \leq 8$$

- A 8
B 16
C 24
D 32
E 48

- 7 Find the sum of the squares of the real roots of the equation

$$x\sqrt{1-x^2} = \frac{1}{3}$$

- A $\frac{1}{9}$
B $\frac{1}{3}$
C $\frac{7}{9}$
D 1
E 2

- 8 A company's annual profit is modelled to decrease by a fixed percentage, $p\%$, at the end of each year.

The total profit predicted for the first two years is 36% of the total profit predicted if the company were to operate forever under these conditions.

What is the value of p ?

- A 20
- B 36
- C 40
- D 60
- E 64
- F 80

- 9 A bag contains 7 gold tokens and 5 silver tokens. The tokens are identical apart from their colour.

A player draws tokens one at a time at random from the bag without replacement.

The player begins with a score of 0.

- Each gold token drawn increases the score by 1 point.
- When a silver token is drawn, the score decreases by 2 points and the game ends immediately.
- If all 7 gold tokens are drawn without drawing a silver token, the game also ends.

What is the probability that the player finishes the game with a score strictly greater than 0?

- A $\frac{35}{396}$
- B $\frac{7}{99}$
- C $\frac{7}{44}$
- D $\frac{5}{22}$
- E $\frac{7}{22}$
- F $\frac{5}{11}$

- 10** A particle undergoes three successive displacements, $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{d}_3$, and returns to its starting position. $\mathbf{d}_1$ has magnitude $\sqrt{3}$ units and a bearing of 090° . $\mathbf{d}_2$ has magnitude 1 unit and a bearing of 180° .

What is the bearing of $\mathbf{d}_3$?

- A** 060°
- B** 120°
- C** 240°
- D** 300°
- E** 330°

- 11** What is the coefficient of x^{24} in the expansion of the expression below?

$$((1 - 2x^2)(1 + 2x^2 + 4x^4 + 8x^6))^5$$

- A** -245760
- B** -40960
- C** -80
- D** 0
- E** 20480
- F** 40960

- 12** The definite integral of a function $f(x)$ over the interval $0 \leq x \leq 6$ is given by $\int_0^6 f(x) \, dx = -12$.
The area of the region bounded by the curve $y = f(x)$, the x -axis, and the lines $x = 0$ and $x = 6$ that lies above the x -axis is 5.
What is the area of the region bounded by the curve, the x -axis, and the lines $x = 0$ and $x = 6$ that lies below the x -axis?

- A** 5
- B** 7
- C** 12
- D** 17
- E** 22

- 13 What is the value of the following sum?

$$\sum_{n=1}^{63} \log_4 \left(\frac{n^2 + n}{n^2 + 2n + 1} \right)$$

- A -6
- B -3
- C 0
- D 3
- E 6

- 14 A straight line with a negative gradient passes through the point (4, 3).

The line forms a triangle with the positive x -axis and positive y -axis.

The line $y = 2x$ divides this triangle into two smaller triangles.

The area of the triangle with a side on the x -axis is three times the area of the triangle with a side on the y -axis.

What is the area of the triangle formed by the straight line and the positive coordinate axes?

- A $\frac{289}{12}$
- B 27
- C $\frac{121}{4}$
- D $\frac{75}{2}$
- E $\frac{289}{6}$

- 15 How many distinct real solutions does the following equation have?

$$\ln(x - 5) + \ln(2 - x) = \ln(2)$$

- A 0
- B 1
- C 2
- D 3
- E 4
- F 7

- 16** A sector of a circle has a fixed perimeter P .
Which expression gives the radius r that maximises the area of the sector?

- A** $\frac{P}{2+\pi}$
B $\frac{P}{4}$
C $\frac{P}{\pi}$
D $\frac{P}{3}$
E $\frac{P}{2}$

- 17** A sequence is defined by $u_1 = 2$ and the recurrence relation

$$u_{n+1} = \ln(e^{u_n-1} - k)$$

for $n \geq 1$, where k is a real constant.

The sequence terminates if $e^{u_n-1} - k \leq 0$.

Find the set of values of k for which the sequence has exactly two strictly positive terms.

- A** $0 \leq k < e - 1$
B $0 \leq k < \frac{e}{e+1}$
C $0 < k \leq e - 1$
D $0 \leq k \leq e - 1$
E $\frac{e^2 - 1}{2} \leq k < e^2 - 1$

- 18** Two rectangles, A and B, have dimensions that depend on a variable x , which changes with time.

The side lengths of rectangle A are $(\log_2(8) + 3x)$ and $(12 - x)$.

The side lengths of rectangle B are $(\log_3(27) + x)$ and $(6 - x)$.

At a certain instant, the perimeter of rectangle A is 42. At this instant, x is decreasing.

What is the value of the ratio $\frac{\text{Area of A}}{\text{Area of B}}$ at this instant, and is this ratio increasing or decreasing?

- A** The ratio is 3 and it is decreasing.
B The ratio is 3 and it is increasing.
C The ratio is 6 and it is decreasing.
D The ratio is 6 and it is increasing.
E The ratio is $\frac{1}{6}$ and it is decreasing.
F The ratio is $\frac{1}{6}$ and it is increasing.

- 19** Find the x -coordinate of the stationary point of the curve with equation:

$$y = \frac{x^2 - 4x}{x^2 - 16}(x + 4)^2$$

- A** -4
- B** -2
- C** 0
- D** 2
- E** 4

- 20** A particle is dropped from a height of 10 m. After each bounce, it reaches a maximum height that is a constant fraction, r , of the maximum height of the previous bounce, where $0 < r < 1$.

The particle comes to rest after travelling a total vertical distance of 50 m.

What is the value of r ?

- A** $r = \frac{1}{2}$
- B** $r = \frac{3}{5}$
- C** $r = \frac{2}{3}$
- D** $r = \frac{5}{7}$
- E** $r = \frac{4}{5}$

- 21** Which of the following statements is/are correct?

1. $2^{\log_3 5} = 5^{\log_3 2}$

2. $\frac{\sqrt{75} - \sqrt{12}}{\sqrt{27}} = 1$

3. $\frac{\log_{10} 8}{\log_{10} 2} = \log_{10} 4$

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 22** Find the sum of all solutions to the equation

$$\ln(-\sin x) + \ln(-\cos x) = -\ln 4$$

in the interval $0 \leq x \leq 2\pi$.

- A** 0
- B** $\frac{\pi}{2}$
- C** π
- D** $\frac{5\pi}{2}$
- E** 3π

- 23** The index, $I(f)$, of a polynomial $f(x)$ is defined by the formula:

$$I(f) = d + s + |f(-1)|$$

where d is the degree of $f(x)$, and s is the number of distinct real stationary points of $f(x)$.

Consider the two polynomials:

$$f_1(x) = x^3 - 3x + 1$$

$$f_2(x) = x^3 - 5$$

What is the value of $I(f_1) + I(f_2)$?

- A** 6
- B** 8
- C** 16
- D** 17
- E** 18
- F** 20

- 24** A rock sample contains a radioactive isotope X and its stable decay product Y. The ratio of the number of nuclei of X to the number of nuclei of Y is 1:3.

The half-life of X is T.

Assuming that there was no Y in the sample initially and that all nuclei of Y were produced by the decay of X, which of the following is the age of the sample?

- A** 1.5 T
- B** 2 T
- C** 3 T
- D** 4 T
- E** T

- 25** The diagram shows a right-angled trapezium with parallel horizontal sides of length $(1 + 3\sqrt{3})$ cm and $(5 + \sqrt{3})$ cm, and a perpendicular height of $(5 - 2\sqrt{3})$ cm.

What is the area of the trapezium?

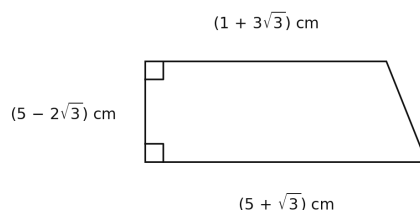


Diagram not to scale

- A** $(3 + 4\sqrt{3}) \text{ cm}^2$
- B** $(6 + 8\sqrt{3}) \text{ cm}^2$
- C** $(27 + 4\sqrt{3}) \text{ cm}^2$
- D** $(21 - 4\sqrt{3}) \text{ cm}^2$
- E** $(3 + 16\sqrt{3}) \text{ cm}^2$
- F** $(9 + 2\sqrt{3}) \text{ cm}^2$

- 26** A sequence is defined by the general term

$$v_n = \frac{9^{n+2}}{3^{2n+1}}$$

for integers $n \geq 1$.

What is the value of the sum of the first 10 terms of the sequence?

- A** $\frac{10}{27}$
 - B** 27
 - C** 30
 - D** 270
 - E** 810
- 27** Find the area of the finite region bounded by the curve with equation $4y + |3x| = 24$ and the x - axis.
- A** 24
 - B** 48
 - C** 96
 - D** 144
 - E** 192
 - F** 288